a = num subproblems @ each level of recursion

b = factor by which problem size is reduced each call

$c_{crit} = \log_a b$

$T(n) = aT\left(\frac{n}{b}\right) + f(n)$

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$f(n)$ = cost of work done outside the recursive calls

$\frac{n}{b^k} = 1 \Rightarrow k = \log_b(n)$

Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asympotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these **25** recurrences in **10** minutes.

Both const = 2

Problem 1-1. $T(n) = 3T(n/2) + n^2$ $a=3, b=2, f(n)=n^2 \to n^{1.585}$ → case 3

$af(n/b) \le cf(n)$ $c_{crit} = \log_2 3 \approx 1.585$ if $f(n) = \log_b a \Rightarrow \Theta(n^{f(n)} \log n)$ $\boxed{T(n) = \Theta(n^2)}$

$n^2 = \log_2 3 \Rightarrow \Theta(n^{n^2} \log n) =$ [illegible]

case 1 **Problem 1-2.** $T(n) = 7T(n/2) + n^2$

$n^{\log_2 7} = n^{1.4}$ n^2

$\Theta(n^{\log_b a}) \to n^{\log 7}$ $\boxed{T(n) = \Theta(n^{\log 7})}$

case 2 **Problem 1-3.** $T(n) = 4T(n/2) + n^2$ $a=4, b=2, f(n)=n^2$

if $f(n) = \log_b a \Rightarrow \Theta(n^{f(n)} \log n)$ $T(n) = \Theta(n^{\log_b a} \log n$

$n^2 = \log_2 4 \Rightarrow \Theta(n^{n^2} \log n) =$ $\boxed{\Theta(n^2 \log n)}$

$2^{n^2 \log n}$

case 3 **Problem 1-4.** $T(n) = 3T(n/4) + n \lg n$

$a = 3$

$b = 4$

$f(n) = n \lg n$

$n^{\log_4 3} = n^{.8} < n \lg n$ $3f(n/4) \le cf(n) \to \frac{3}{4}$ $\boxed{\Theta(n \log n)}$

$\Theta(n^{\log_4 3} \log^2(n)) = \Theta(n \lg n)$

case 1 **Problem 1-5.** $T(n) = 4T(n/2) + \lg n$

$a = 4$

$b = 2$

$f(n) = \log n$

$n^{\log_2 4} = n^2 > \log n$

$\boxed{\Theta(n^2)}$

✗ **Problem 1-6.** $T(n) = T(n-1) + n$

Does not Apply

$$T(n) = \begin{cases} \Theta \\ \Theta(n^{\log_b a} \log^{k+1} n) \end{cases}$$

case 2 **Problem 1-7.** $T(n) = 4T(n/2) + n^2 \lg n$

$a = 4$
$b = 2$
$f(n) = n^2 \log n$

$n^{\log_b a} \lg^{k+1} n$

$n^{\log_2 4} = n^2$

$\Theta(n^2 \lg^2 n)$

case 1 **Problem 1-8.** $T(n) = 5T(n/2) + n^2 \lg n$

$a = 5$
$b = 2$

$n^{\log 5} \to n^{2.3} > n^2 \log n$

$\Theta(n^{\log 5})$

Problem 1-9. $T(n) = 3T(n/3) + n/\lg n$

$a = 3$ $n^{\log_3 3} = n$
$b = 3$
$f(n) = \frac{n}{\log n}$ ← negative

Does not apply

Problem 1-10. $T(n) = 2T(n/4) + c$

case 1 $a = 2$ $n^{\log_4 2} = n^{(.5)}$ c
$b = 4$
$f(n) = c$

$\Theta(n^{\log_b a}) = n^{(.5)} \to \Theta(\sqrt{n})$

case 2 **Problem 1-11.** $T(n) = T(n/4) + \lg n$

$n^{\log_4 1}$

$n^{\log_4 1} \log^2 n \to \log^2 n$

$n^{\log_b a} \lg^{k+1} n$

$\Theta(\lg^2 n)$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

Does not apply

case 1 **Problem 1-13.** $T(n) = 2T(n/4) + \lg n$

$a = 2$
$b = 4$ $> n^{\log_4 2} = n^{.5}$
$f(n) = \lg n$

$\Theta(\sqrt{n})$

case 2 **Problem 1-14.** $T(n) = 3T(n/3) + n \lg n$ $\to \Theta(n^{\log_b a} \lg^{k+1} n)$

$a = 3$
$b = 3$

$n^{\log_3 3} = n^1$

$\Theta(n \lg^2 n)$

Problem 1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

Does not apply

case 2 **Problem 1-16.** $T(n) = 2T(n/4) + \sqrt{n}$

$n^{\log_4 2} = n^{.5}$ $\boxed{\Theta(\sqrt{n}\log n)}$

case 3 **Problem 1-17.** $T(n) = 2T(n/4) + n^{0.51}$

$n^{\log_4 2} = n^{.5}$ $f(n) = n^{0.51}$

$\boxed{\Theta(n^{0.51})}$

case 3 **Problem 1-18.** $T(n) = 16T(n/4) + n!$

$n^{\log_4 16} = n^2$ $\boxed{\Theta(n!)}$

case 1 **Problem 1-19.** $T(n) = 3T(n/2) + n$

$n^{\log_2 3} = n^{1.6}$

$\boxed{\Theta(n^{\log_2 3})}$

case 1 **Problem 1-20.** $T(n) = 4T(n/2) + cn \to n$

$n^{\log_2 4} = n^2$

$\boxed{\Theta(n^2)}$

case 2 **Problem 1-21.** $T(n) = 3T(n/3) + n/2$

$\log_3 3 \to n$

$\boxed{\Theta(n\log n)}$

case 1 **Problem 1-22.** $T(n) = 4T(n/2) + n/\lg n$

$\hookrightarrow n^{\log_2 4} = n^2$

$\boxed{\Theta(n^2)}$

case 3 **Problem 1-23.** $T(n) = 7T(n/3) + n^2$

$n^{\log_3 7} = n^{1.8}$ → $\boxed{\Theta(n^2)}$

case 3 **Problem 1-24.** $T(n) = 8T(n/3) + 2^n$

$n^{\log_3 8} = n^{1.9}$

$\boxed{\Theta(2^n)}$

case 1 **Problem 1-25.** $T(n) = 16T(n/4) + n$

$n^{\log_4 16} = n^2$ ↗

$\boxed{\Theta(n^2)}$